

TIME SERIES MODELS

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Outline

- 1 AR ($p=1$) process
- 2 MA process
- 3 Stationarity
- 4 Wold's Representation Theorem
- 5 Seasonality and Seasonal Adjustment
- 6 VAR
- 7 Local Projections

AR ($p=1$) process

- Let x_t be a time series. Under some conditions, x_t can be modelled as an auto-regressive process with a lag of one.
- $x_{t+1} = \mu + \rho x_t + \epsilon_{t+1}$ where ϵ_{t+1} distributes iid $N(0, \sigma^2)$
- $E_t x_{t+1} = \mu + \rho x_t$
- $E x_t = \frac{\mu}{1-\rho}$
- $V_t x_{t+1} = V_t \epsilon_{t+1}$
- $V x_{t+1} = \rho^2 V x_t + V \epsilon_{t+1} + \text{Cov}(x_t, \epsilon_{t+1})$ (This cov is zero, why?)
- $V x_{t+1} = V \epsilon_{t+1} / (1 - \rho^2)$

AR (p=1) process

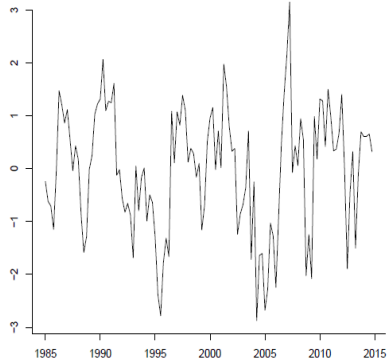
- Let x_t be an AR process.
- $x_{t+1} = \mu + \rho x_t + \epsilon_{t+1}$
- $x_{t+1} = \mu + \rho(\mu + \rho x_{t-1} + \epsilon_t) + \epsilon_{t+1}$
- $x_{t+1} = \mu + \rho(\mu + \rho(\mu + \rho x_{t-2} + \epsilon_{t-1})) + \epsilon_t + \epsilon_{t+1}$
- $x_{t+1} = \mu + \rho\mu + \dots + \rho^t\mu + \rho^{t+1}x_0 + \rho^t\epsilon_0 + \dots + \rho\epsilon_t + \epsilon_{t+1}$
- $x_{t+1} = \frac{\mu}{1-\rho} + \sum \rho^k \epsilon_{t+1-k}$

AR (p=1) process

- $x_{t+1} = \frac{\mu}{1-\rho} + \sum \rho^k \epsilon_{t+1-k}$
- Based on this expression, let's obtain its expected value, variance, and covariance.
- $x_{t+1} = \frac{\mu}{1-\rho} + \sum \rho^k \epsilon_{t+1-k}$
- $E x_{t+1} = \frac{\mu}{1-\rho} + \sum E \rho^k \epsilon_{t+1-k} = \frac{\mu}{1-\rho}$
- $V x_{t+1} = \sum \rho^{2k} V(\epsilon_{t+1-k}) = V(\epsilon_{t+1-k}) \sum \rho^{2k} = V(\epsilon_{t+1-k}) / (1 - \rho^2)$
- All along, we have made two assumptions.

AR ($p=1$) process

Figure 1: AR Processes



(a) AR(1) with $\alpha[1] = 0.5$



(b) AR(1) with $\alpha = 0.95$

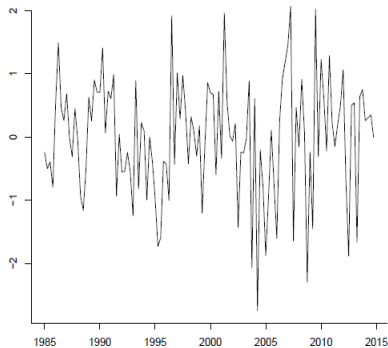
Source: Hansen (2021).

MA process

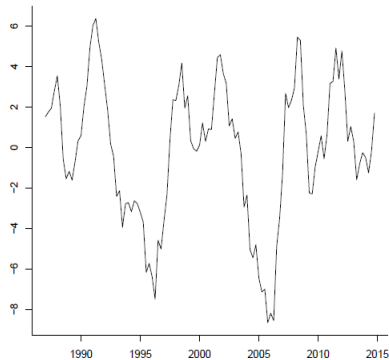
- $x_{t+1} = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}$ (MA(1))
- $x_{t+1} = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$ (MA(q))
- $x_{t+1} = \frac{\mu}{1-\rho} + \sum \rho^k \epsilon_{t+1-k}$ (MA(∞))
- ARMA (p,q)

MA process

Figure 2: MA Processes



(a) White Noise



(b) MA(8)

Source: Hansen (2021).

Stationarity I

- Let x_t be a time series
- Weak stationarity or co-variance stationarity
 - ▶ The mean $\mu = E(x_t)$ and covariance matrix $\text{var}(x_t) = E[(x_t - \mu)(x_t - \mu)']$ are finite and are independent of t
 - ▶ $\text{cov}(x_t, x_{t+h}) = E[(x_t - \mu)(x_{t+h} - \mu)']$ are independent of t for all h
- Strictly stationarity
 - ▶ The joint distribution of $(x_t, \dots, x_{t+l})$ is independent of t for all l
 - ▶ $f(x_t, x_{t+h}) = f(x_{t+j}, x_{t+j+h})$ for all t, j , and h
 - ▶ The distribution only depends on the periods separating x_t and x_{t+h}

Stationarity II

- Let x_t be a AR(1) $x_{t+1} = \frac{\mu}{1-\rho} + \sum \rho^k \epsilon_{t+1-k}$
- Weak stationarity
 - ▶ $E(x_t) = E(x_{t+h})$ for all t and h
 - ▶ $cov(x_t, x_{t+h}) = cov(\sum \rho^k \epsilon_{t-k}, \sum \rho^k \epsilon_{t+h-k}) = \rho^h \sigma^2 / (1 - \rho^2)$. It only depends on h
- Strong stationarity
 - ▶ $f(x_t, x_{t+h}) = f(x_{t+j}, x_{t+j+h})$ for all t, j , and h
 - ▶ The distribution only depends on the periods separating x_t and x_{t+h}
 - ▶ If shocks are normal, one needs to check the first two moments

Wold's Representation Theorem

- Let x_t be a weak stationary time series, then X_t can be written as:
- $X_t = LD_t + \sum \alpha_i \epsilon_{t-i}$

Where

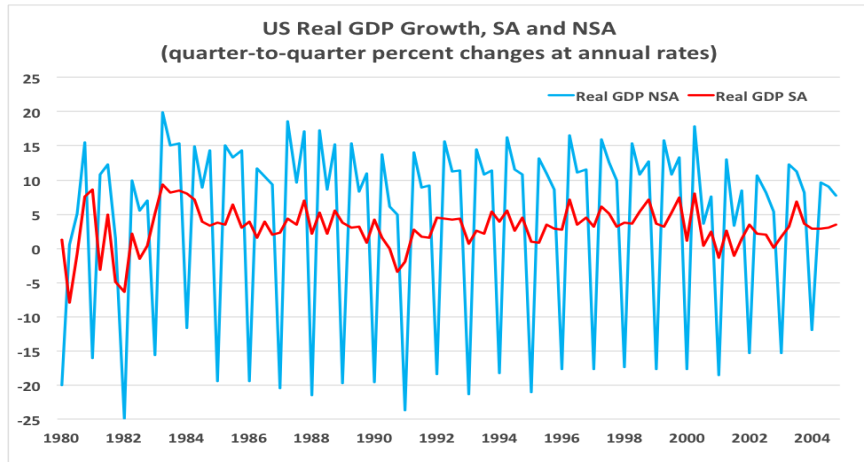
- LD is a linear and deterministic process
- ϵ_t is an iid process
- Benefits and Perils

Seasonality and Seasonal Adjustment

- Let x_t be a time series. Seasonality refers to a variation of the time series that takes place at regular intervals such as monthly or quarterly
- For instance, sales increase every December. Ice-cream sales in summer
- Seasonal component should not be confused with a cyclical one, in which the variation occurs at varying intervals
- There are methods to adjust a time series for seasonality
- Prominently: X12-ARIMA, SEATS
- There are other related adjustments, such as accounting for holidays

Seasonality and Seasonal Adjustment

Figure 3: US real GDP growth (q/q percentage at annual rates)



Source: <https://www.moneyandbanking.com/>.

- Let x_t be a vector time series. Under some conditions, x_t can be modelled as a **vector** auto-regressive process with a lag of one
- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where ϵ_{t+1} is iid $N(0, \Omega)$
- $E_t x_{t+1} = \mu + Ax_t + E_t \epsilon_{t+1} = \mu + Ax_t$
- $E_t x_{t+2} = \mu + A(\mu + Ax_t + E_t \epsilon_{t+1}) + E_t(\epsilon_{t+2}) = \mu + A\mu + A^2 x_t$
- $E_t x_{t+k} = \sum A^k \mu + A^k x_t$
- $E x_{t+1} = (I - A)^{-1} \mu = \sum A^k \mu = Q \sum D^k Q' \mu$
- $A = QDQ'$ is a SVD, where Q is orthogonal matrix and D is a diagonal matrix with the singular values of A . (SVD. Singular Value Decomposition)

- Let x_t be a vector time series. Under some conditions, x_t can be modelled as a **vector** auto-regressive process with a lag of one
- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where ϵ_{t+1} is iid $N(0, \Omega)$
- $Ex_{t+1} = (I - A)^{-1}\mu = \sum A^k \mu = Q \sum D^k Q' \mu$
- $V_t x_{t+1} = V_t \epsilon_{t+1} = \Omega$
- $x_{t+1} = (I - A)^{-1}\mu + \sum A^k \epsilon_{t+1-k}$
- $Vx_{t+1} = \sum A^k \Omega A'^k = \sum QD^k Q' \Omega (QD^k Q')'$
- $Vx_{t+1} = Q \sum D^k Q' \sigma^2 I QD^k Q' = \sigma^2 Q \sum D^{2k} Q'$ if $\Omega = \sigma^2 I$

Impulse Response Function

Univariate time series. The coefficients of the moving average representation

$$Y_t = b(L)e_t$$

$$Y_t = \sum_{j=0}^{\infty} b_j e_{t-j}$$

$$Y_t = b_0 e_t + b_1 e_{t-1} + b_2 e_{t-1} + \dots$$

In linear models the impulse response function is defined as the change in Y_{t+j} due to a shock at time t . This is

$$\frac{\partial Y_{t+j}}{\partial e_t} = b_j$$

Impulse Response Function

In applied multivariate time series is the impulse response function (IRF) which is defined as the change in Y_t due to a change in an innovation or shock e_t .

The multivariate Wold representation

$$Y_t = \mu + \sum_{j=0}^{\infty} \Theta_j e_{t-j}$$

Thus,

$$\text{IRF}(h) = \frac{\partial \text{Projection}_t[Y_{t+h}]}{\partial e_t}$$

$$\text{Projection}_t[Y_{t+h}] = \mu + \sum_{j=h}^{\infty} \Theta_j e_{t+h-j} = \mu + \sum_{j=0}^{\infty} \Theta_{h+j} e_{t-j}$$

$\text{IRF}(h) = \Theta_h$, the h^{th} moving average coefficient matrix.

VAR III Structural/Reduced forms

- $x'_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where $\epsilon_{t+1} = \Sigma z_{t+1}$ and $\Sigma\Sigma' = \Omega$, z has a variance covariance matrix I .
- Estimation OLS, line by line.
- $$\begin{pmatrix} \Sigma_{1,1} & \Sigma_{1,2} \\ \Sigma_{2,1} & \Sigma_{2,2} \end{pmatrix} \begin{pmatrix} \Sigma_{1,1} & \Sigma_{2,1} \\ \Sigma_{1,2} & \Sigma_{2,2} \end{pmatrix} = \begin{pmatrix} \Omega_{1,1} & \Omega_{1,2} \\ \Omega_{2,1} & \Omega_{2,2} \end{pmatrix}$$
- $\epsilon_{t+1,1}$ is a statistical shock.
- $z_{t+1,1}$ is an economic shock.
- $\Sigma^{-1}x_{t+1} = \Sigma^{-1}\mu + \Sigma^{-1}Ax_t + z_{t+1}$ (Structural form)
- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ (Reduced form)

VAR IV Identification

- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where $\epsilon_{t+1} = \Sigma z_{t+1}$ and $\Sigma\Sigma' = \Omega$
- Estimation OLS, line by line.
- $$\begin{pmatrix} \Sigma_{1,1} & \Sigma_{1,2} \\ \Sigma_{2,1} & \Sigma_{2,2} \end{pmatrix} \begin{pmatrix} \Sigma_{1,1} & \Sigma_{2,1} \\ \Sigma_{1,2} & \Sigma_{2,2} \end{pmatrix} = \begin{pmatrix} \Omega_{1,1} & \Omega_{1,2} \\ \Omega_{2,1} & \Omega_{2,2} \end{pmatrix}$$
- $\Sigma_{1,1}\Sigma_{1,1} + \Sigma_{1,2}\Sigma_{1,2} = \Omega_{1,1}$
- $\Sigma_{1,1}\Sigma_{2,1} + \Sigma_{1,2}\Sigma_{2,2} = \Omega_{1,2}$
- $\Sigma_{2,1}\Sigma_{1,1} + \Sigma_{2,2}\Sigma_{1,2} = \Omega_{2,1}$
- $\Sigma_{2,1}\Sigma_{2,1} + \Sigma_{2,2}\Sigma_{2,2} = \Omega_{2,2}$
- We have 4 unknowns ($\Sigma_{1,1}$, $\Sigma_{1,2}$, $\Sigma_{2,1}$ and $\Sigma_{2,2}$) and 3 equations (why?).

VAR V Identification (SR, Recursive, Sims)

- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where $\epsilon_{t+1} = \Sigma z_{t+1}$ and $\Sigma\Sigma' = \Omega$
- **Assuming** Σ is lower triangular.
- $$\begin{pmatrix} \Sigma_{1,1} & 0 \\ \Sigma_{2,1} & \Sigma_{2,2} \end{pmatrix} \begin{pmatrix} \Sigma_{1,1} & \Sigma_{2,1} \\ 0 & \Sigma_{2,2} \end{pmatrix} = \begin{pmatrix} \Omega_{1,1} & \Omega_{1,2} \\ \Omega_{2,1} & \Omega_{2,2} \end{pmatrix}$$
- $\Sigma_{1,1}\Sigma_{1,1} = \Omega_{1,1}$
- $\Sigma_{1,1}\Sigma_{2,1} = \Omega_{1,2}$
- $\Sigma_{2,1}\Sigma_{1,1} = \Omega_{2,1}$
- $\Sigma_{2,1}\Sigma_{2,1} + \Sigma_{2,2}\Sigma_{2,2} = \Omega_{2,2}$
- Only a shock on the first variable has a **contemporaneous** effect on the second one.
- We have 3 unknowns and 3 equations. Cholesky decomposition of Ω

VAR VI Identification (LR, Blanchard-Q)

- $x_{t+1} = \mu + Ax_t + \epsilon_{t+1}$ where $\epsilon_{t+1} = \Sigma z_{t+1}$ and $\Sigma\Sigma' = \Omega$
- $\Sigma, A\Sigma, A^2\Sigma, \dots$
- $L = (I - A)^{-1}\Sigma$
- $LL' = (I - A)^{-1}\Sigma\Sigma'(I - A)^{-1'}$
- One assumes that L is lower triangular.
- $\begin{pmatrix} L_{1,1} & 0 \\ L_{2,1} & L_{2,2} \end{pmatrix} \begin{pmatrix} L_{1,1} & L_{2,1} \\ 0 & L_{2,2} \end{pmatrix}$
- Only a shock on the first variable has a **long run** effect on the second one.
- Cholesky decomposition of $(I - A)^{-1}\Sigma\Sigma'(I - A)^{-1'}$

VAR VII Identification

- Identification through heteroskedasticity ([Rigobon, 2013](#); [Lütkepohl, Meitz, Netsunajev, and Saikkonen, 2018](#))
- External instruments ([Mertens and Ravn, 2013](#); [Ramey, 2016](#))
- Sign and zero restrictions ([Uhlig, 2005](#); [Arias, Rubio-Ramirez and Waggoner, 2018](#))

VAR - Sign and Zero Restrictions

The SVAR can be written as

$$\mathbf{y}'_t \mathbf{A}_0 = \mathbf{x}'_t \mathbf{A}_+ + \varepsilon'_t;$$

where

$$\mathbf{A}'_+ \equiv \begin{bmatrix} \mathbf{A}'_1 & \cdots & \mathbf{A}'_p & \mathbf{D}' \end{bmatrix}, \quad \mathbf{x}'_t \equiv \begin{bmatrix} \mathbf{y}'_{t-1} & \cdots & \mathbf{y}'_{t-p} & \mathbf{w}'_t \end{bmatrix},$$

so that its reduced form representation is

$$\mathbf{y}'_t = \mathbf{x}'_t \mathbf{B} + \mathbf{u}'_t; \tag{1}$$

where $\mathbf{B} \equiv \mathbf{A}_+ (\mathbf{A}_0)^{-1}$, $\mathbf{u}'_t \equiv \varepsilon'_t (\mathbf{A}_0)^{-1}$, and $E[\mathbf{u}_t \mathbf{u}'_t] = \mathbf{\Sigma} = (\mathbf{A}_0 \mathbf{A}'_0)^{-1}$.

VAR - Sign and Zero Restrictions

Then the coefficients $\mathbf{B}$ are estimated from (1) by ordinary least squares (OLS), so that

$$\hat{\mathbf{B}} = \left[\sum_{t=1}^T \mathbf{y}'_t \mathbf{x}_t \right] \left[\sum_{t=1}^T \mathbf{x}'_t \mathbf{x}_t \right]^{-1},$$

and $\mathbf{\Sigma}$ is recovered through the estimated residuals $\hat{\mathbf{u}}'_t = \mathbf{y}'_t - \mathbf{x}'_t \hat{\mathbf{B}}$.

VAR - Sign and Zero Restrictions

Given the estimation of the reduced form, now we turn to the identification of structural shocks.

Partial identification, since in general we are only interested in some of the shocks. The algorithm for the estimation is as follows

- 1 Draw $(\mathbf{B}, \Sigma)$ independently from the posterior $NIW(B, \Sigma, Q)$ distribution
- 2 Draw Q independently from the uniform distribution
- 3 Keep $(\mathbf{A}_0, \mathbf{A}_+)_k = f_h^{-1}(B, \Sigma, Q)$ that satisfies the zero and sign restrictions of the shock, and compute the impulse responses (f_h are densities over the orthogonal reduced-form parameterization that induce densities over the structural parameterization)
- 4 Return to Step 1 until the required number of draws has been obtained

VAR - Sign and Zero Restrictions

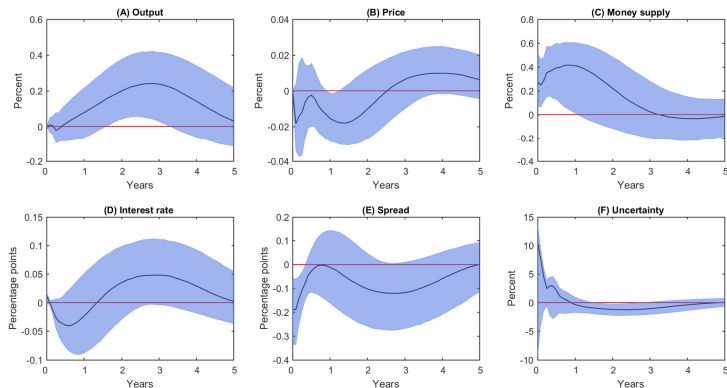
Table 1: Identification restrictions for a QE policy shock in the U.S.

Variable	QE shock ($h = 0$)	QE shock ($h = 1, 2$)
Uncertainty	?	?
Spread	—	—
Money supply	+	+
Interest rate	0	0
Price	0	?
Output	0	?

Note: ? = left unconstrained.

VAR - Sign and Zero Restrictions

Figure 4: Impulse responses of U.S. variables after a QE policy shock



Note: median value (solid line) and 68% bands (shadow area).

VAR - Sign and Zero Restrictions

Let the first shock be the monetary policy shock (Arias et al. 2018). Thus, the first equation

$$\mathbf{y}'_t \mathbf{a}_{0,1} = \sum_{i=1}^p \mathbf{y}'_{t-i} \mathbf{a}_{i,1} + \varepsilon_{1t}, \quad (2)$$

is the monetary policy equation, where ε_{1t} denotes the first entry of ε_t , $\mathbf{a}_{i,1}$ denotes the first column of $\mathbf{A}_i$ for $0 \leq i \leq p$, and $a_{i,jk}$ denotes the (j, k) entry of $\mathbf{A}_i$.

Thus, in order to restrict the systematic component of the monetary policy, we need to restrict $\mathbf{a}_{i,1}$.

The interest rate is the monetary policy instrument and the contemporaneous reaction of the interest rate to output and prices is usually positive.

VAR - Sign and Zero Restrictions

To implement this identification strategy, the reduced-form VAR consists of six endogenous variables that are ordered as follows: output, y_t ; price, p_t ; commodity prices, e_t ; total reserves, cd_t ; nonborrowed reserves, cf_t ; and a policy interest rate, i_t . Thus, equation (2), abstracting from lag variables, can be written as

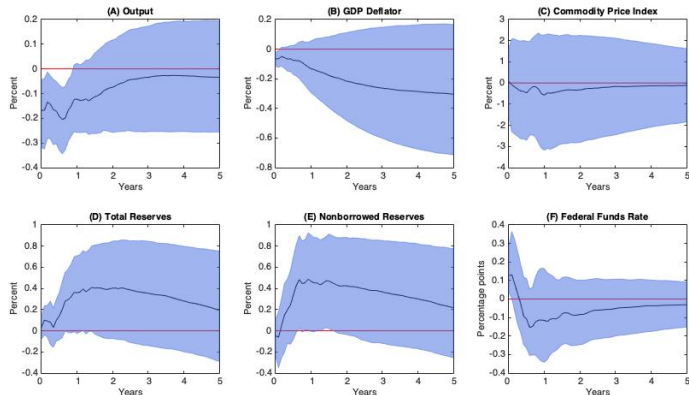
$$i_t = \psi_y y_t + \psi_p p_t + \psi_e e_t + \psi_{cd} cd_t + \psi_{cf} cf_t + \sigma \varepsilon_{1t}; \quad (3)$$

restrictions on the systemic component of the monetary policy rule imply $\psi_{cd} = \psi_{cf} = 0$, $\psi_y, \psi_p > 0$ and ψ_e remains unrestricted.

Note that $\psi_y = -a_{0,61}^{-1} a_{0,11}$, $\psi_p = -a_{0,61}^{-1} a_{0,21}$, $\psi_e = -a_{0,61}^{-1} a_{0,31}$, $\psi_{cd} = -a_{0,61}^{-1} a_{0,41}$, $\psi_{cf} = -a_{0,61}^{-1} a_{0,51}$ and $\sigma = -a_{0,61}^{-1}$.

VAR - Sign and Zero Restrictions

Figure 5: Impulse responses of U.S. variables after a monetary policy shock



Note: median value (solid line) and 68% bands (shadow area).

Local Projections

As in [Hansen \(2022b\)](#), in prediction it is useful to consider models where the dependent variable is dated multiple periods ahead of the right-hand-side variables.

$$Y_{t+h} = b_0 + B_1 Y_1 + \dots + B_p Y_{t-p+1} + u_t,$$

where the integer $h \geq 1$ is the horizon. [Hansen \(2022b\)](#) argues that there is a relationship between a VAR model and the corresponding h -step predictive VAR model.

If Y_t is a VAR(p) process then its h -step predictive regression is a predictive VAR(p) with u_t a MA($h-1$) process and $B_1 = \Theta_h = IRF(h)$, where $IRF(h)$ is the impulse response function which is defined as the change in Y_t due to a change in an innovation or shock for each horizon h .

Local Projections

The impulse response can be estimated by a least squares predictive regression ([Jordà, 2005](#)).

The key is that $B_1 = \Theta_h$, the leading coefficient matrix in the h -step predictive regression.

The method is as follows. For each horizon h estimate a predictive regression to obtain the leading coefficient matrix estimator $\hat{B}_1$. The estimator is $\widehat{IRF}(h) = \hat{B}_1$ and is known as the local projection estimator.

In addition, the local projection impulse response estimator is asymptotically normal.

References

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